

Generating and Collecting Sports Data

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HES 3000-002, Sport Performance Analytics

Case Study 1

Share:

- Their sport concern
- Their analytical question
- Who would use the answer
- Their ideal evidence
- Their feasible data
- Their biggest remaining uncertainty

- Data submission MUST be an excel or csv (comma separated variable) file.
 - Knowing where the data exists is very different from having the data
 - Data submission accepted with no late penalty until 10 Sep, 11:59 PM

Analytics Project

- 1. Identify a meaningful sport problem.**
- 2. Develop a focused analytical question.**
3. Determine what evidence is needed.
4. Find and evaluate feasible data.
5. Conduct an analysis that matches the question.
6. Interpret the results for the intended user.

Before the next checkpoint, your group should know what each row in your proposed dataset represents, who or what generated the data, and whether the variables needed for your question are actually present.

Where do the numbers come from?

HOW IS SPORTS DATA COLLECTED?

Observational

- Scout with notebook, coach's tactical board, video coding
- Strengths: Contextual, can capture nuanced behaviors
- Limitations: Schema determines possibilities, subjective, time-intensive



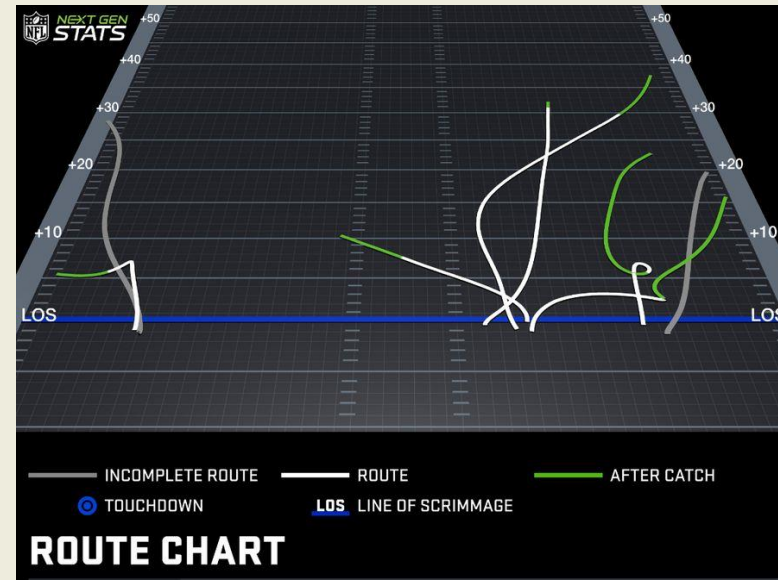
Optical tracking systems

- Camera-based position tracking (Second Spectrum, Hawk-Eye)
- Strengths: Spatial precision, non-invasive, every player tracked
- Limitations: Expensive, venue-dependent, line-of-sight issues



Wearable technology

- GPS units, accelerometers, heart rate monitors, IMUs
- Strengths: Continuous data, objective physiological metrics
- Limitations: Only captures what sensors detect, requires athlete buy-in



Laboratory Equipment

- Force plates, motion capture, metabolic carts, isokinetic testing
- Strengths: Gold-standard precision, controlled conditions
- Limitations: Artificial environment, doesn't capture competition reality



Who collects Sports data?

League/Competition Providers

- Leagues may provide standardize data across all teams/matches
 - Examples: NBA stats, FIFA match data
- Pros: Consistency, benchmarking
- Cons: One-size-fits-all, slow to innovate



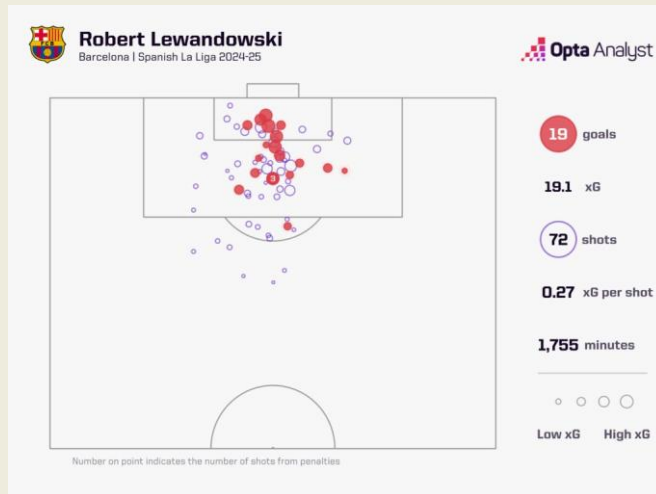
Teams (Internal data)

- Teams often try to collect and analyze data to gain a competitive advantage
 - Examples: Proprietary load monitoring, custom tactical metrics
- Pros: Tailored to needs, flexibility, privacy
- Cons: No external benchmarks, siloed innovation, inaccessible for outsiders



Third-Party Vendors

- Some businesses sell data collection technology and data analysis software
 - Opta, Stats Perform, Catapult, Kinexon, etc.
- Pros: Expertise, scale, cross-sport learning
- Cons: Black box algorithms, vendor lock-in, no competitive advantage



Academic/Research Institutions

- Research institutions prioritize methodological rigor.
- Pros: Reliable methods, scientific collaboration
- Cons: Limited samples, specific contexts, slow dissemination

Primary Data Collection

- You collect data for a specific question or use case
- Pros: Full control over what and how data are collected
- Cons: Time and resource intensive, access can be a challenge



Where does Sports Data Come From?

Human-recorded data

- Examples:
 - Scouting grades
 - Notational analysis
 - Coach observations
 - Injury or participation records
 - Event tagging

Instrument-generated data

- Examples:
 - GPS or local positioning systems
 - Optical tracking
 - Force plates
 - Heart-rate monitors
 - Timing systems

Event → Dataset

- A player performs a jump → force plates record force → software identifies takeoff → an equation estimates jump height → one jump-height value appears in a table.

Summary & Looking Ahead

- The three domains describe where sport analytics is applied.
- Descriptive, diagnostic, predictive, and prescriptive questions describe what the analysis is intended to accomplish.
 - These purposes are connected, but not every analysis requires all four.
- The most valuable analysis is the one that provides credible evidence for a meaningful decision.
- Next: Turning practitioner concerns into clear, testable analytical questions